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MIL-C-0050072C(MU)
19 July 1967
USED IN LIEU OF
MIL-C-50072B
30 September 1964

MILITARY SPECIFICATION

COVER, VOICEMITTER-OUTLET VALVE, C13

This limited coordination Military specification has been prepared by Edgewood Arsenal based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-C-50072B. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 This specification describes one type of molded cover for the voicemitter-outlet valve of the M17 protective mask.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

FED-STD-601	-	Rubber: Sampling and Testing.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 4240

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DRAWINGS

US ARMY MUNITIONS COMMAND

D5-2-954 - Cover, Voicemitter-Outlet Valve, C13.

PUBLICATIONS

EDGEWOOD ARSENAL DIRECTIVE

No. 618 - Ozone Resistance Test of Natural Rubber Material.

(Copies of specifications, standards, drawings, and publications required by the suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials.

3.1.1 Rubber compound. The rubber compound shall be non-blooming, except for slight wax-type bloom, and shall contain not less than 65 percent, by volume, of new natural rubber. The compound shall be free from lead or mineral sulfates.

3.1.2 Antioxidants and accelerators. The rubber compound shall contain only approved antioxidants and accelerators.

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3.1.3 Sun-checking retardant. The rubber compound shall contain a sun-checking retardant.

3.1.4 Copper and manganese inhibitor. The rubber compound shall contain a copper and manganese inhibitor.

3.1.5 Coloring materials. The rubber compound shall be colored by carbon black, as required.

3.2 Manufacture. The cover shall be manufactured in one piece by the conventional process of molding under heat and pressure, in accordance with Drawing D5-2-954, in molds approved by Edgewood Arsenal.

3.2.1 Repair of flaws. Covers with small flaws may be repaired and accepted, provided there is no change in the formulation or surface continuity.

3.2.2 Test slabs. Test slabs, 6 by 6 inches, by 0.075 ± 0.01 inches, shall be made from stock representative of the formulation used for the preproduction sample and from each sublot during regular production of covers. The slabs shall have a cure equivalent to that of the regular production covers (see 6.3).

3.3 Physical properties. Physical properties shall conform to the requirements shown in table I when tested as specified therein. Specimens for tests shall be prepared from test slabs (see 3.2.2) as specified in 4.3.4.1.

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TABLE I. Physical properties

Property	Unaged	After heat aging	Test paragraph
Average tensile strength, psi, min	3000	2550	4.3.4.2(a)
Average ultimate elongation, %, min	600		4.3.4.2(a)
Percent increase (max)		20	
Percent decrease (max)		25	
Average tensile stress, psi at 200% elongation	250 to 450	250 to 450	4.3.4.2(a)
Individual specimens, psi, min	240	240	
Average tear resistance, lb/in, min			4.3.4.2(a)
Die A or B	300	210	
Die C	225	165	
Average tension set at 75% ultimate elongation, %, max	20		4.3.4.2(a)
Ozone resistance, min	80		4.3.4.2(b)
Flexibility			4.3.4.2(c)
Units at 75° ± 5°F., min	7.5		
Units retained at -25° ± 5°F., min	4.8		
*Puncture resistance, psi, min	9.5		4.3.4.2(d)

*To be performed on preproduction sample only.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of covers and test slabs shall be produced in accordance with this specification for examination and tests (see 4.2). Any change in formulation, operating conditions, or molds shall require a new preproduction sample.

3.5 Workmanship. The cover's inner and outer surfaces shall be smooth and free from imperfections, such as pits, bumps, porous areas, cracks, checks, flash, abrasions, or blisters. The rubber shall be compact and free from holes, bubbles, and foreign matter. Ridges of mold parting lines on outer surfaces shall not be considered as flash when closely trimmed. Rips, tears, or punctures shall be considered as damage.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of 25 covers, with a minimum of 2 covers from each mold cavity, and 39 test slabs shall be manufactured using the same methods, materials, equipment, and rubber formulation as will be used during regular production. (See 6.2)

4.2.2 Inspection.

4.2.2.1 For examination. The sample covers shall be inspected for all requirements of the drawings and this specification.

4.2.2.2 For tests. Test slabs produced in accordance with 4.2.1 (see 3.2.2) shall be tested in accordance with 4.3.4.

4.2.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.2.2.1 and 4.2.2.2 to be acceptable. The supplier shall obtain written approval from the contracting officer before proceeding with regular production.

4.3 Regular production.

4.3.1 Lotting. A regular production lot shall consist of the covers produced in sequence during one month, as indicated by the dots in the medallion on the cover.

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A sublot shall consist of the covers produced in not more than one work week from one batch of material (see 6.6). A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or the use of an alternate approved formulation. Acceptably repaired covers from any accepted parent lot may be incorporated into and assume the lot number of current production.

4.3.2 Sampling.

4.3.2.1 For examination. Sampling shall be conducted in accordance with MIL-STD-105.

4.3.2.2 For destructive tests. Thirty-nine test slabs (see 3.2.2) shall be submitted with each sublot for testing.

4.3.3 Inspection

4.3.3.1 For examination. The sample covers from regular production lots shall be examined in accordance with the classification of defects and with MIL-STD-105.

4.3.3.2 For destructive tests. Sample test slabs shall be tested in accordance with 4.3.4. The acceptance number is 0.

4.3.3.3 Classification of defects.

(a) Cover, voicemitter-outlet valve (Dwg D5-2-954).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Workmanship (see 3.5)
102	Fit incorrect
103	Rivet-hole diameter or alignment incorrect
104	Anti-tear slit incorrect or incorrectly located
105	Anti-tear slit not clean cut or free from nicks or flash
Minor:	AQL 2.5 percent defective
201	Thickness incorrect
202	Marking missing, illegible, or incorrect

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4.3.4 Tests.

4.3.4.1 Test specimens. Specimens shall not be buffed or stretched before testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tensile strength, ultimate elongation, and tensile stress testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Six specimens, two from each of three test slabs, shall be taken, without aging, for tension-set testing. Twenty specimens, two from each of 10 test slabs, shall be taken for tear-resistance testing. Ten specimens shall be tested before aging and 10 shall be tested after aging. Twelve specimens, two from each of six test slabs, shall be taken for ozone-resistance testing. Eight specimens, two from each of four test slabs, shall be taken for flexibility testing. Six specimens, one from each of six test slabs, shall be taken for puncture-resistance testing. The test specimens selected for heat aging shall be aged in an air oven at a temperature of 158° ± 1.8°F. for a continuous period of 168 ± 1 hours or at a temperature of 212° ± 1.8°F. for 24 ± 1 continuous hours, as specified in FED-STD-601 (see 6.7).

4.3.4.2 Physical properties test.

(a) Tensile strength, ultimate elongation, tensile stress, tear resistance, and tension set. Tensile strength, ultimate elongation, tensile stress, tear resistance, and tension set shall be determined as specified in FED-STD-601. The angle-type or crescent type die shall be used for tear-resistance testing.

(b) Ozone resistance. Ozone resistance shall be determined in accordance with Directive No. 618.

(c) Flexibility. Flexibility shall be determined with Q37 Natural Rubber Flexibility Tester. Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 75° ± 5°F. and four specimens shall be tested at -25° ± 5°F. after being conditioned for 72 hours at each temperature.

(d) Puncture resistance. Resistance to hydrostatic pressure shall be determined with Q38 Natural Rubber Puncture Tester.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The covers shall be packaged to provide adequate protection against deterioration, and damage from the supply source to the first receiving activity for immediate use.

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5.2 Packing, level C. Covers, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

5.4 Repair part. When this item is procured for storage and issue as a repair part, preservation, packaging, packing and marking shall be as specified on the repair part packaging data sheet which is identified by the Federal stock number for the repair part.

6. NOTES

6.1 Intended use. This specification describes a protective cover intended to be used on the C29R1 Voicemitter-Outlet Valve Assembly.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preproduction.

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Equivalent cure. Slabs and covers of the same stock and gage are considered to have equivalent cures or the same state of vulcanization when:

(a) Their free-sulfur contents do not vary more than 10 percent.
(b) Their average tensile-stress values at any given elongation (other than ultimate elongation) do not vary more than $\pm$ 2.5 percent.
(c) Their curing times can be equated in the following equation:

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$$*\left(\frac{T_1 - T_2}{10}\right)$$

$$t_2 = t_1 \times 1.6$$

where: t_1 = minutes of cure at temperature T_1,

t_2 = minutes of cure at temperature T_2,

T_1 = temperature (°F.) of cure for time t_1, and

T_2 = temperature (°F.) of cure for time t_2.

*The quantity $\left(\frac{T_1 - T_2}{10}\right)$ is exponential to the constant 1.6.

For convenience, the nomograph in figure 1 may be used instead of the mathematical computation time-temperature relationships in determining equivalent cure.

6.4 Major change. The term applies to any change in formula, processing, or curing of currently approved rubber stocks. The major change should be considered affected when one or more of the following conditions occur:

(a) Components of formulae, are varied in excess of the limitations stated in the manufacturers' tolerance (see 6.5).
(b) The addition or deletion of one or more components of formulae.
(c) Substitutes for one or more components of formulae without advance approval of Edgewood Arsenal.
(d) Deviations from the submitted and approved factory mold cure exceeding ± 5°F. or -10°F., even though equivalent cure is attained.
(e) Use of mold lubricants, dusting powders, or wet sprays not used in preparation or cure of the original samples submitted for approval.

6.5 Manufacturer's tolerances. The manufacturer should submit to Edgewood Arsenal the variations in amounts used, or tolerances, for accelerators, plasticizers, and softeners. These tolerances are permitted to enable the manufacturer to produce stock which will consistently comply with this specification and which will process satisfactorily in the plant. Components of formulae may be submitted on the basis of percentage of stock or number of parts of rubber, and the manufacturer's tolerances recorded in the same units.

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EQUIVALENT CURE TIME FOR NATURAL RUBBER

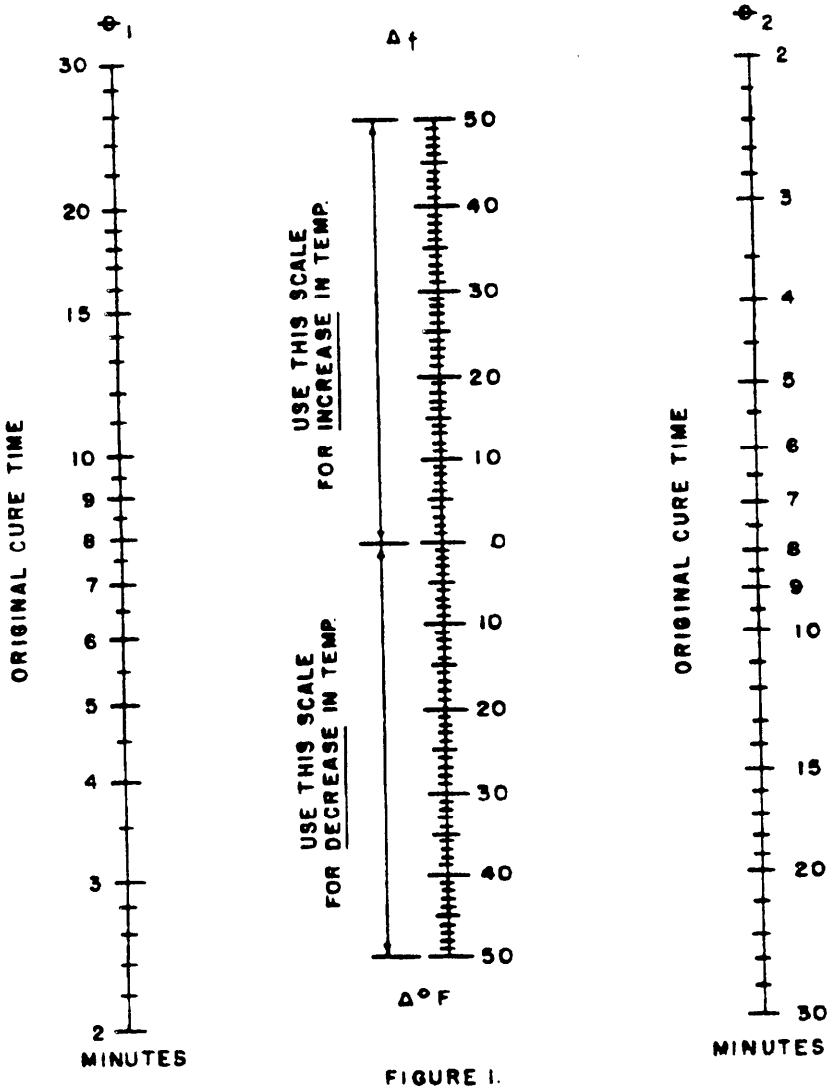


FIGURE I.

USING A STRAIGHT EDGE, CONNECT THE ORIGINAL CURE TIME ON Θ_1 SCALE WITH THE CHANGE OF CURE TEMPERATURE ON Δt SCALE. EXTEND THE LINE UNTIL IT INTERSECTS Θ_2 SCALE AT THE EQUIVALENT CURE TIME.

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6.6 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.7 Aging. Aging at 212° ± 1.8°F. for a continuous period of 24 hours is slightly more severe on rubber than aging at 158° ± 1.8°F. for 168 consecutive hours; therefore, specimens that fail to meet requirements when aged at the higher temperature may be discarded and replaced by new specimens aged at 158° ± 1.8°F. for 168 consecutive hours.

6.8 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 21, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodian:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 4240-A319

International interest (see 6.8)

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-C-0050072C(MU)
NOTICE 1
28 September 1995

MILITARY SPECIFICATION

COVER, VOICEMITTER-OUTLET VALVE, C13

This notice should be filed in front of MIL-C-0050072C(MU) dated 19 July 1967.

MIL-C-0050072C(MU) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

Project No. 4240-A142

AMSC N/A

FSC 4240

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